

# AUTOM-04 LAB: Terraform for Dynatrace

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**Series:** AUTOM — Dynatrace Automation | **Notebook:** 4 of 8 (LAB) | **Created:** April 2026 | **Last Updated:** 04/25/2026

## Overview

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Hands-on lab for installing the Dynatrace Terraform provider, configuring authentication, creating and managing Dynatrace resources as code, importing existing configuration, and integrating into a CI/CD pipeline with GitHub Actions.

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# Prerequisites

| Requirement           | Details                                         |
|-----------------------|-------------------------------------------------|
| Completed             | AUTOM-04: Terraform Provider (lecture notebook) |
| Dynatrace Environment | SaaS tenant (Gen3)                              |
| Terraform CLI         | Version 1.0+ installed                          |
| Authentication        | API Token and/or OAuth Client credentials       |
| Permissions           | Settings write, IAM admin (for Section 5)       |
| Git                   | Git CLI and a GitHub account (for Section 9)    |

## 1. Install Terraform

Terraform is a single binary. Download it from [terraform.io](https://terraform.io) or use a package manager.

### macOS (Homebrew):

```
brew tap hashicorp/tap
brew install hashicorp/tap/terraform
```

### Linux (APT — Debian/Ubuntu):

```
wget -O- https://apt.releases.hashicorp.com/gpg | sudo gpg --dearmor -o
/usr/share/keyrings/hashicorp-archive-keyring.gpg
echo "deb [signed-by=/usr/share/keyrings/hashicorp-archive-keyring.gpg]
https://apt.releases.hashicorp.com $(lsb_release -cs) main" | sudo tee
/etc/apt/sources.list.d/hashicorp.list
sudo apt-get update &&& sudo apt-get install terraform
```

### Windows (Chocolatey):

```
choco install terraform
```

### Verify the installation:

```
terraform version
# Expected output: Terraform v1.x.x
```

**Note:** The Dynatrace Terraform provider requires Terraform 1.0 or later. Any 1.x release will work.

## 2. Configure the Dynatrace Provider

Create a project directory and a `main.tf` file:

```
mkdir dynatrace-terraform && cd dynatrace-terraform
touch main.tf
```

Add the provider block to `main.tf`:

```
terraform {
  required_providers {
    dynatrace = {
      source = "dynatrace-oss/dynatrace"
      version = "~> 1.93"
    }
  }
}
```

### Authentication Method 1: API Token (Simplest)

Best for getting started. The API Token covers Settings 2.0, Synthetics, and SLOs.

```
provider "dynatrace" {
  dt_env_url    = "https://<env-id>.live.dynatrace.com"
  dt_api_token = var.dt_api_token
}
```

## Authentication Method 2: OAuth Client Credentials (Recommended for Automation)

Required for IAM resources and Gen3 platform resources (workflows, documents, segments).

```
provider "dynatrace" {  
  dt_env_url      = "https://&lt;env-id&gt;.live.dynatrace.com"  
  dt_client_id    = var.dt_client_id  
  dt_client_secret = var.dt_client_secret  
  dt_account_id   = var.dt_account_id  
}
```

**Important (v1.88.0):** OAuth-based authentication **cannot manage synthetic monitors or SLO definitions** as of provider v1.88.0. Use an API Token for those resources, or configure both tokens together.

## Authentication Method 3: Environment Variables

Keep credentials out of `.tf` files entirely:

```
export DYNATRACE_ENV_URL="https://&lt;env-id&gt;.live.dynatrace.com"  
export DYNATRACE_API_TOKEN="dt0c01.xxx..."
```

With environment variables set, the provider block needs no arguments:

```
provider "dynatrace" {}
```

## Initialize the Provider

Run `terraform init` to download the provider plugin:

```
terraform init  
# Initializing provider plugins...  
# - Installing dynatrace-oss/dynatrace v1.93.x...  
# Terraform has been successfully initialized!
```

### 3. Create Your First Resource — Alerting Profile

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Add an alerting profile resource to `main.tf` :

```
resource "dynatrace_alerting" "production" {
  name = "Production Alerts"
  rules {
    rule {
      severity_level = "AVAILABILITY"
      delay_in_minutes = 0
    }
    rule {
      severity_level = "ERROR"
      delay_in_minutes = 5
    }
  }
}
```

#### Preview the Change

```
terraform plan
```

Expected output:

```
Terraform will perform the following actions:

# dynatrace_alerting.production will be created
+ resource "dynatrace_alerting" "production" {
+   id      = (known after apply)
+   name    = "Production Alerts"
+   rules {
+     rule {
+       severity_level = "AVAILABILITY"
+       delay_in_minutes = 0
+     }
+     rule {
+       severity_level = "ERROR"
+       delay_in_minutes = 5
+     }
+   }
+ }

Plan: 1 to add, 0 to change, 0 to destroy.
```

## Apply the Change

```
terraform apply
# Type "yes" when prompted to confirm
```

## Verify the Resource

```
terraform show
```

Confirm the alerting profile exists in your Dynatrace tenant under **Settings > Alerting > Alerting profiles**.

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## 4. Create Settings 2.0 Resources

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The `dynatrace_generic_setting` resource can manage any Settings 2.0 schema. This is useful when the provider does not have a dedicated resource type for a particular setting.

### Example: Kubernetes Metadata Enrichment Rule

```
resource "dynatrace_generic_setting" "k8s_enrichment" {
  schema_id = "builtin:kubernetes.generic.metadata.enrichment"
  scope     = "environment"
  value = jsonencode({
    enabled = true
    rules = [{
      sourceAttribute = "namespace-label"
      sourceKey       = "team"
      targetAttribute = "dt.security_context"
    }]
  })
}
```

## Key Attributes

| Attribute | Description                                                                                                              |
|-----------|--------------------------------------------------------------------------------------------------------------------------|
| schema_id | The Settings 2.0 schema identifier (e.g., <code>builtin:kubernetes.generic.metadata.enrichment</code> )                  |
| scope     | Where the setting applies: <code>environment</code> , <code>HOST-xxx</code> , <code>KUBERNETES_CLUSTER-xxx</code> , etc. |
| value     | JSON-encoded configuration object matching the schema definition                                                         |

## Finding Schema IDs

Navigate to **Settings** in the Dynatrace UI. The schema ID appears in the URL when you open any setting:

```
https://&lt;env-  
id&gt;.apps.dynatrace.com/ui/settings/builtin:kubernetes.generic.metadata.enr  
ichment
```

```

-----
This is the schema_id

```

Alternatively, use the Settings API:

```
curl -H "Authorization: Api-Token $DT_API_TOKEN" \
  "https://&lt;env-id&gt;.live.dynatrace.com/api/v2/settings/schemas" | jq
'.items[].schemaId'
```

## 5. Create IAM Resources (OAuth Required)

IAM resources (groups, policies, bindings) require OAuth client credentials. API tokens cannot manage IAM.

## Create an IAM Group

```
resource "dynatrace_iam_group" "sre_team" {  
  name = "SRE Team"  
}
```

## Create an IAM Policy

```
resource "dynatrace_iam_policy" "sre_policy" {  
  name           = "SRE Full Access"  
  statement_query = "ALLOW environment:roles:viewer;"  
  tags           = ["sre"]  
}
```

## Bind the Policy to the Group

```
resource "dynatrace_iam_policy_bindings_v2" "sre_binding" {  
  group_id   = dynatrace_iam_group.sre_team.id  
  account_id = var.dt_account_id  
  policies   = [dynatrace_iam_policy.sre_policy.id]  
}
```

## Apply

```
terraform plan  
terraform apply
```

**Note:** The `account_id` variable is required for IAM policy bindings. This is your Dynatrace account UUID, found under **Account Management > Account settings**.



## 6. Import Existing Resources

If you already have Dynatrace resources configured manually, you can bring them under Terraform management without recreating them.

## Step 1: Add a Resource Block

Create an empty resource block in your `.tf` file:

```
resource "dynatrace_alerting" "existing_profile" {
  # Configuration will be filled after import
}
```

## Step 2: Find the Resource ID

Locate the resource ID from the Dynatrace UI or API. For alerting profiles, it appears in the URL:

[illegible]

## Step 3: Import

```
terraform import dynatrace_alerting.existing_profile abc12345-def6-7890-abcd-ef1234567890
```

## Step 4: Generate the HCL

After importing, extract the configuration from state:

```
terraform show -no-color > imported.tf
```

Review and clean up the generated HCL. Remove read-only attributes (like `id` ) and format the code.

## Common Import Targets

| Resource Type                          | ID Source                                   |
|----------------------------------------|---------------------------------------------|
| <code>dynatrace_alerting</code>        | Settings UI URL or API                      |
| <code>dynatrace_dashboard</code>       | Dashboard URL ( <code>id</code> parameter)  |
| <code>dynatrace_slo</code>             | SLO list page or API                        |
| <code>dynatrace_generic_setting</code> | Settings API ( <code>objectId</code> field) |

**Tip:** After import, run `terraform plan` to verify the imported state matches the live configuration. Any differences indicate drift that you should reconcile in the `.tf` file.

## 7. Variables and Multi-Environment

### Define Variables

Create `variables.tf` :

```
variable "dt_env_url" {
  type      = string
  description = "Dynatrace environment URL"
}

variable "dt_api_token" {
  type      = string
  sensitive = true
  description = "Dynatrace API token"
}

variable "environment" {
  type      = string
  default   = "dev"
  description = "Target environment name (dev, staging, production)"
}
```

## Supply Values with tfvars

Create `terraform.tfvars` (add to `.gitignore` — never commit credentials):

```
dt_env_url    = "https://&lt;env-id&gt;.live.dynatrace.com"
dt_api_token  = "dt0c01.xxx..."
environment   = "production"
```

## Approach 1: Workspace-Based Multi-Environment

Terraform workspaces maintain separate state files per environment using the same configuration:

```
# Create workspaces
terraform workspace new dev
terraform workspace new staging
terraform workspace new production

# Switch between environments
terraform workspace select production

# Use workspace name in configuration
# terraform.workspace returns the current workspace name
```

Reference the workspace in your resources:

```
resource "dynatrace_alerting" "alerts" {
  name = "${terraform.workspace}-alerts"
  # ...
}
```

## Approach 2: Directory-Based Multi-Environment

For larger teams, use separate directories with shared modules:

```
dynatrace-terraform/  
  modules/  
    alerting/  
      main.tf  
      variables.tf  
  environments/  
    dev/  
      main.tf          # calls module with dev values  
      terraform.tfvars  
    staging/  
      main.tf  
      terraform.tfvars  
    production/  
      main.tf  
      terraform.tfvars
```

Each environment directory runs independently with its own state and variables.

---

## 8. State Management

---

Terraform state tracks the mapping between your `.tf` files and real Dynatrace resources. By default, state is stored locally in `terraform.tfstate`.

### Local State (Default)

Works for single-operator setups. The state file is created automatically after `terraform apply`.

**Warning:** Never commit `terraform.tfstate` to version control. It may contain sensitive values. Add it to `.gitignore`.

### Remote State — S3 Backend

For team use, store state in a shared remote backend:

```

terraform {
  backend "s3" {
    bucket      = "my-terraform-state"
    key         = "dynatrace/terraform.tfstate"
    region      = "us-east-1"
    dynamodb_table = "terraform-locks"
    encrypt     = true
  }
}

```

| Attribute      | Purpose                                                              |
|----------------|----------------------------------------------------------------------|
| bucket         | S3 bucket for state storage                                          |
| key            | Path within the bucket                                               |
| dynamodb_table | DynamoDB table for state locking (prevents concurrent modifications) |
| encrypt        | Encrypt state at rest                                                |

## Remote State — Azure Blob

```

terraform {
  backend "azurerm" {
    resource_group_name = "rg-terraform"
    storage_account_name = "tfstateaccount"
    container_name       = "tfstate"
    key                  = "dynatrace.terraform.tfstate"
  }
}

```

## Remote State — Terraform Cloud

```

terraform {
  cloud {
    organization = "my-org"
    workspaces {
      name = "dynatrace-production"
    }
  }
}

```

## Inspecting State

```
# List all resources in state
terraform state list

# Show details of a specific resource
terraform state show dynatrace_alerting.production

# Remove a resource from state (without destroying it)
terraform state rm dynatrace_alerting.production
```

---

## 9. GitHub Actions CI/CD Pipeline

Automate Terraform deployments with GitHub Actions. The workflow runs `plan` on pull requests and `apply` on merge to main.

### Workflow File

Create `.github/workflows/terraform-deploy.yml` :

```
name: Deploy Dynatrace Terraform

on:
  pull_request:
    paths: ["terraform/**"]
  push:
    branches: [main]
    paths: ["terraform/**"]

jobs:
  plan:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v4
      - uses: hashicorp/setup-terraform@v3

      - name: Terraform Init
        run: terraform init
        working-directory: terraform/

      - name: Terraform Plan
        run: terraform plan -no-color
        working-directory: terraform/
        env:
          DYNATRACE_ENV_URL: ${ secrets.DT_ENV_URL }
          DYNATRACE_API_TOKEN: ${ secrets.DT_API_TOKEN }

  apply:
    if: github.ref == 'refs/heads/main' && github.event_name == 'push'
    needs: plan
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v4
      - uses: hashicorp/setup-terraform@v3

      - name: Terraform Init
        run: terraform init
        working-directory: terraform/

      - name: Terraform Apply
        run: terraform apply -auto-approve
        working-directory: terraform/
        env:
          DYNATRACE_ENV_URL: ${ secrets.DT_ENV_URL }
          DYNATRACE_API_TOKEN: ${ secrets.DT_API_TOKEN }
```

## GitHub Secrets

Store credentials in GitHub repository secrets (**Settings > Secrets and variables > Actions**):

| Secret Name  | Value                               |
|--------------|-------------------------------------|
| DT_ENV_URL   | https://<env-id>.live.dynatrace.com |
| DT_API_TOKEN | Your Dynatrace API token            |

## Adding Approval Gates

For production deployments, require manual approval before `terraform apply` :

1. Create a GitHub Environment named `production` under **Settings > Environments**
2. Enable **Required reviewers** and add approvers
3. Reference the environment in the `apply` job:

```
apply:
  environment: production    # Requires approval
  needs: plan
  # ...
```

---

## 10. Drift Detection

Drift occurs when someone changes a Dynatrace resource manually (through the UI or API) after it was deployed via Terraform.

### Detect Drift

```
# Run plan with detailed exit code
terraform plan -detailed-exitcode

# Exit codes:
# 0 = No changes (no drift)
# 1 = Error
# 2 = Changes detected (drift found)
```



## Scheduled Drift Detection

Add a scheduled GitHub Actions workflow to check for drift daily:

```
name: Drift Detection

on:
  schedule:
    - cron: "0 8 * * 1-5"    # Weekdays at 8 AM UTC

jobs:
  detect-drift:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v4
      - uses: hashicorp/setup-terraform@v3
      - run: terraform init
        working-directory: terraform/
      - name: Check for drift
        id: drift
        run: terraform plan -detailed-exitcode
        working-directory: terraform/
        continue-on-error: true
      - name: Alert on drift
        if: steps.drift.outcome == 'failure'
        run: echo "Drift detected! Review terraform plan output."
```

## Handling Drift

| Scenario                      | Action                                                                           |
|-------------------------------|----------------------------------------------------------------------------------|
| Manual change was intentional | Update <code>.tf</code> to match, or <code>terraform import</code> the new state |
| Manual change was accidental  | Run <code>terraform apply</code> to restore the desired state                    |
| State is stale                | Run <code>terraform refresh</code> to update state from the live tenant          |

# 11. Summary and Checklist

## Deployment Checklist

- [ ] Terraform 1.0+ installed and verified
- [ ] Provider configured with correct authentication
- [ ] `terraform init` downloads the Dynatrace provider
- [ ] Resources defined in `.tf` files, not configured manually
- [ ] `terraform.tfvars` and `terraform.tfstate` in `.gitignore`
- [ ] Remote state backend configured for team use
- [ ] CI/CD pipeline runs `plan` on PRs, `apply` on merge
- [ ] Drift detection scheduled

## Key Takeaways

| Concept           | Key Point                                                           |
|-------------------|---------------------------------------------------------------------|
| Provider Setup    | Pin version with <code>~&gt; 1.93</code> , configure authentication |
| Authentication    | API Token for Synthetics/SLOs, OAuth for IAM and Gen3               |
| Plan Before Apply | Always review <code>terraform plan</code> output before applying    |
| State Management  | Use remote backends for team collaboration                          |
| Import            | Bring existing resources under Terraform control without recreation |
| CI/CD             | Automate with GitHub Actions; plan on PR, apply on merge            |
| Drift Detection   | Schedule regular checks with <code>plan -detailed-exitcode</code>   |

## Monaco vs Terraform — When to Use Which

| Scenario                    | Recommended Tool           |
|-----------------------------|----------------------------|
| Quick config export/import  | Monaco                     |
| Multi-environment promotion | Either                     |
| IAM policy management       | Terraform (OAuth required) |

| Scenario                               | Recommended Tool              |
|----------------------------------------|-------------------------------|
| Multi-cloud infrastructure + Dynatrace | Terraform                     |
| Drift detection and state tracking     | Terraform                     |
| Team with existing Terraform expertise | Terraform                     |
| Team new to IaC                        | Monaco (lower learning curve) |

## References

| Resource               | URL                                                                                                                                                           |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Terraform Registry     | <a href="https://registry.terraform.io/providers/dynatrace-oss/dynatrace">registry.terraform.io/providers/dynatrace-oss/dynatrace</a>                         |
| Provider Documentation | <a href="https://registry.terraform.io/providers/dynatrace-oss/dynatrace/latest/docs">registry.terraform.io/providers/dynatrace-oss/dynatrace/latest/docs</a> |
| Provider GitHub        | <a href="https://github.com/dynatrace-oss/terraform-provider-dynatrace">github.com/dynatrace-oss/terraform-provider-dynatrace</a>                             |
| Terraform Install      | <a href="https://developer.hashicorp.com/terraform/install">developer.hashicorp.com/terraform/install</a>                                                     |
| Dynatrace IAM Docs     | <a href="https://docs.dynatrace.com/docs/manage/identity-access-management">docs.dynatrace.com/docs/manage/identity-access-management</a>                     |

*Return to AUTOM-04: Terraform Provider for conceptual coverage, or continue to AUTOM-05: Dynatrace Workflows for event-driven automation.*

## Community Resources

| Repository                                              | Description                                                                                          |
|---------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| <a href="#">terraform-provider-dynatrace</a>            | Official provider (v1.93.0, 180 releases) with export capability                                     |
| <a href="#">dynatrace-configuration-as-code-samples</a> | 10 starter templates in <code>basic-templates-terraform</code> , plus modules, IAM, and DQL examples |
| <a href="#">terraform_modules</a>                       | Reusable module pattern for synthetic monitors                                                       |
| <a href="#">terraform_dql_example</a>                   | DQL as a Terraform data source for dynamic config generation                                         |

| Repository                                | Description                                                    |
|-------------------------------------------|----------------------------------------------------------------|
| <a href="#">terraform_team_onboarding</a> | IAM policies, groups, and Azure Entra ID for team provisioning |
| <a href="#">iam_tf_sample</a>             | IAM policies, Grail buckets, OpenPipelines, and segments       |

**Tip:** Use the provider export feature ( `./terraform-provider-dynatrace -export -env <url> -token <token>` ) to generate `.tf` files from an existing tenant -- the fastest path to Terraform-managed configuration.

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*This notebook was AI-generated from community-submitted and publicly available sources. This notebook series is not officially supported by Dynatrace. Always verify information against official Dynatrace documentation.*